

Test Plan

AI Student Evaluator

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Document

Attribute	Details
Document Name	AI Student Evaluator – Test Plan
Version	1.0 (Final Submission)
Author	Utkarsh Yadav
Audience	Competition Judges, Software Engineers, QA Engineers
Backend	Java 21, Maven, Javalin
Frontend	React, Recharts, jsPDF
AI Engine	Google Gemini 2.5 Flash

Revision History

Version	Date	Author	Description
0.1	04 July 2026	Utkarsh Yadav	Final Draft
0.5	04 July 2026	Utkarsh Yadav	Architecture Defined
1.0	04 July 2026	Utkarsh Yadav	Competition Submission

1. Introduction

The purpose of this document is to provide evidence that the **AI Student Evaluator** has been systematically tested across functional, integration, performance, and AI-specific scenarios.

The testing validates not only the correctness of the application but also the quality and consistency of AI-generated evaluations.

2. Testing Objectives

The primary objectives of testing were to:

- Verify correctness of each application module.
- Validate AI-generated evaluations.
- Ensure reliable parsing of WhatsApp conversations.
- Verify dashboard generation.
- Validate JSON export.
- Ensure PDF report generation.
- Identify and document defects.
- Demonstrate consistent behaviour across multiple scenarios.

3. Scope of Testing

Included

- Mentor summary ingestion
- WhatsApp parsing
- Student grouping
- Gemini AI evaluation
- Dashboard generation
- JSON export
- PDF export

Excluded

- Authentication
- Cloud deployment
- LMS integration
- Multimedia inputs

4. Test Environment

Component	Details
Operating System	Windows 11
Backend	Java 21
Build Tool	Maven
Frontend	React
AI Model	Google Gemini 2.5 Flash
JSON Library	Jackson

5. Acceptance Criteria

- All planned functional test cases have been executed successfully.
- All Critical (P0) and High (P1) defects have been resolved before sign-off.
- AI evaluation generates valid scores, strengths, missing concepts, mentor recommendations, and learning goals for every student.
- The dashboard correctly displays student evaluations, rankings, analytics, charts, and detailed performance insights.
- JSON and PDF reports are generated successfully without data loss or formatting issues.
- End-to-end evaluation workflow executes successfully from mentor summary ingestion to dashboard and report generation.
- Functional, AI accuracy, negative, edge case, and integration testing have been completed.
- Testing evidence, execution summary, and defect validation reports have been documented.
- The application meets the defined functional requirements and is approved for final submission.

5.1 QA Sign-off Checklist

Item	Status
Functional Testing Completed	☒
AI Accuracy Validation Completed	☒
Negative & Edge Case Testing Completed	☒
End-to-End Testing Completed	☒
Dashboard Validation Completed	☒
PDF & JSON Export Verified	☒
Critical Defects Closed	☒
Test Evidence Prepared	☒
QA Sign-off Approved	☒

5.2 Entry Criteria

Testing began after:

- Backend implementation completed
- Dashboard integrated
- Gemini API configured
- Parser validated

- JSON export implemented

5.3 Exit Criteria

Testing was considered complete when:

- Critical functional scenarios passed
- Dashboard rendered successfully
- AI evaluations were generated consistently
- PDF export succeeded
- No critical defects remained open

6. Test Data

The AI Student Evaluator was validated using carefully designed test datasets that simulate realistic classroom discussions. The datasets were created to evaluate both the application's functionality and the AI model's ability to accurately assess varying levels of student understanding.

The test data covers a wide range of learning scenarios, including complete understanding, partial understanding, incorrect understanding, missing concepts, and invalid inputs.

6.1 Test Data Source

The test data used during validation was derived from real classroom sessions to ensure that the evaluation scenarios closely reflected practical mentoring environments.

Source	Purpose
Tactiq.ai Meeting Transcript	Used to generate the mentor reference summary from the recorded classroom session.
WhatsApp Group Chat	Used as the primary source of student understanding and discussion.
AI Student Evaluator	Used to generate student evaluations, class analytics, dashboard data, and PDF reports.

7. Testing Strategy

The project followed a layered testing strategy.

Functional Testing

To verify that every feature performs according to its intended business functionality.

Before validating AI quality, it is essential to confirm that the application itself correctly processes inputs, executes business logic, and generates outputs.

The following modules were validated:

- Mentor Summary Reader
- WhatsApp Chat Parser
- Student Grouping Logic
- AI Evaluation Engine
- Response Parser
- Class Analytics Generator
- Dashboard Generation
- JSON Export
- PDF Report Generation

Integration Testing

Even if individual modules function correctly, failures can still occur when modules exchange data.

Integration testing validates that all components work together as a complete system.

The following integrations were validated:

- Mentor Summary → Prompt Builder
- WhatsApp Parser → Student Objects
- Prompt Builder → Gemini API
- Gemini API → Response Parser
- Response Parser → Student Evaluation Objects
- Student Evaluations → Analytics Engine
- Analytics Engine → Dashboard Generator
- Dashboard Generator → React Dashboard
- Dashboard → PDF Export

Negative Testing

Applications should remain stable even when users provide invalid inputs or when external dependencies fail.

Negative testing verifies the robustness of the application under such conditions.

Scenarios included:

- Empty mentor summary
- Empty WhatsApp chat
- Invalid chat format
- Unsupported message structure
- Gemini API failures
- Invalid AI response
- Malformed JSON
- Network interruptions
- PDF export failure

Edge Case Testing

Applications should remain stable even when users provide invalid inputs or when external dependencies fail.

Negative testing verifies the robustness of the application under such conditions.

Examples include:

- Single student in the class
- Large class containing many students
- Student sending only one message
- Student sending a very large number of messages
- Duplicate student names
- Very long mentor summaries
- Messages containing emojis or special characters
- Students summarizing only a subset of the topics
- Extremely short responses
- Repeated concepts across multiple messages

AI Accuracy Validation

Unlike traditional software, AI-generated responses are probabilistic rather than deterministic. Therefore, validation must focus on the quality and relevance of the output instead of expecting identical responses.

This phase confirms that the AI evaluates students in a manner that aligns with the mentor's expectations.

The evaluation specifically verified that:

- High-quality summaries received higher scores.
- Partial understanding resulted in moderate scores.
- Incorrect understanding resulted in lower scores.
- Missing concepts were correctly identified.
- Recommendations aligned with the identified knowledge gaps.
- Feedback remained relevant to the student's understanding.

Performance Testing

The system should remain responsive even when evaluating larger classroom discussions.

Performance testing measures whether the application can process data efficiently while maintaining acceptable response times.

The following metrics were observed:

- Time required to parse WhatsApp chats.
- AI response time per student.
- Total evaluation duration.
- Dashboard generation time.
- PDF generation time.

End-to-End Testing

The ultimate goal is to verify that the complete application workflow functions correctly from the user's perspective. The complete workflow consisted of:

1. Loading the mentor summary.
2. Pasting the WhatsApp chat.
3. Parsing student discussions.
4. Evaluating each student using Google Gemini.

5. Generating structured evaluation results.
6. Calculating class analytics.
7. Rendering the React dashboard.
8. Exporting the final PDF report.

8. Test Scenarios

8.1 Functional Testing

TC ID	Scenario	Expected Result
TC-F-001	Load valid mentor summary	Summary loaded successfully
TC-F-002	Paste valid WhatsApp chat	Chat accepted and processed
TC-F-003	Parse multiple students	Students grouped correctly
TC-F-004	Merge multiple messages from same student	Messages combined correctly
TC-F-005	Generate AI prompt	Prompt contains mentor summary and student response
TC-F-006	Evaluate complete understanding	High score generated
TC-F-007	Evaluate partial understanding	Moderate score generated
TC-F-008	Parse AI JSON	StudentEvaluation object created
TC-F-009	Generate class analytics	Average score and rankings calculated
TC-F-010	Display dashboard	Dashboard renders successfully
TC-F-011	Export dashboard PDF	PDF generated successfully
TC-F-012	Export dashboard JSON	JSON exported successfully

8.2 Negative Scenarios

TC ID	Scenario	Expected Result
TC-N-001	Empty mentor summary	Validation error displayed

TC-N-002	Empty WhatsApp chat	Evaluation prevented
TC-N-003	Gemini API unavailable	Error handled gracefully
TC-N-004	Invalid AI JSON	Parsing error handled
TC-N-005	PDF export failure	Dashboard remains available

8.3 Edge Case Testing

TC ID	Scenario	Expected Result
TC-E-001	Single student	Evaluation completed successfully
TC-E-002	Student sends many messages	Messages merged correctly
TC-E-003	Very long mentor summary	Summary processed successfully
TC-E-004	Duplicate student names	Parsing remains consistent
TC-E-005	Large classroom	Evaluation completes successfully

8.4 Integration Testing

TC ID	Scenario	Expected Result
TC-I-001	Parser → AI Service	Prompt generated correctly
TC-I-002	AI Service → Response Parser	JSON parsed successfully
TC-I-003	Analytics → Dashboard	Dashboard populated correctly
TC-I-004	Dashboard → PDF Export	PDF generated with correct data

8.5 Performance Testing

TC ID	Scenario	Expected Result
TC-P-001	Evaluate small class	Response within acceptable time
TC-P-002	Evaluate larger class	Application remains stable

8.6 AI Accuracy Validation

TC ID	Scenario	Expected Result
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TC-AI-001	Student covers all concepts	High score with minimal missing concepts
TC-AI-002	Student covers partial concepts	Moderate score with identified gaps
TC-AI-003	Student includes incorrect concepts	Low score with corrective feedback
TC-AI-004	Two similar summaries	Consistent scoring and recommendations
TC-AI-005	Student misses a key topic	Missing concept correctly identified
TC-AI-006	Recommendation quality	Recommendation aligns with identified learning gaps

9. AI Accuracy Validation

9.1 Objective

The AI Student Evaluator relies on Google Gemini 2.5 Flash to perform semantic evaluation rather than keyword matching. Therefore, validating the correctness, consistency, and relevance of AI-generated evaluations is a critical part of the testing process.

The objective of AI Accuracy Validation is to ensure that the generated scores, strengths, missing concepts, and recommendations accurately reflect the student's understanding of the mentor's reference summary.

Unlike deterministic software testing, AI outputs may vary slightly between executions. Therefore, validation focuses on the quality and consistency of the evaluation rather than expecting identical responses.

9.2 Validation Criteria

Validation Criteria	Description
Concept Coverage	Were the concepts explained by the student correctly identified?
Score Appropriateness	Does the assigned score match the quality of the student's understanding?
Missing Concepts	Are the omitted topics correctly highlighted?
Strength Identification	Are the student's strengths accurately identified?
Recommendation Quality	Is the recommendation relevant and actionable?
Feedback Consistency	Does the feedback align with the assigned score and identified gaps?

9.3 AI Accuracy Test Cases

TC ID	Scenario	Expected Behaviour	Result
TC-AI-001	Student covers all major concepts	High score with minimal missing concepts	Pass
TC-AI-002	Student explains only part of the session	Moderate score with identified gaps	Pass
TC-AI-003	Student provides incorrect understanding	Low score with corrective feedback	Pass

TC-AI-004	Two similar summaries	Similar evaluation quality and consistent scoring	Pass
TC-AI-005	Student omits an important concept	Missing concept identified in evaluation	Pass
TC-AI-006	Recommendation quality	Recommendation aligns with learning gaps	Pass

10.Automation Plan

10.1 Automation Framework

Layer	Framework
Backend Testing	JUnit 5
API/Service Testing	REST Assured
UI Testing	Playwright / Selenium
Build Tool	Maven
CI/CD	GitHub Actions

10.2 Backend Automation

10.2.1 Components Planned for Automation

- Mentor Summary Reader
- WhatsApp Chat Parser
- Prompt Generation
- Response Parser
- Student Evaluation Service
- Class Analytics Service
- Dashboard Generation
- PDF Export
- JSON Export

10.2.2 Test Cases Planned for Automation

No.	Automation Scenario
1	Verify mentor summary is loaded successfully
2	Verify WhatsApp chat is parsed correctly
3	Verify student messages are grouped accurately
4	Verify AI response is parsed into StudentEvaluation objects
5	Verify analytics (average, highest, lowest score) are calculated correctly

6	Verify DashboardData JSON is generated correctly
7	Verify PDF report is generated successfully

10.3 UI Automation

Framework: Playwright / Selenium

10.3.1 Functionalities Planned for Automation

- Dashboard loading
- Summary cards
- Student search
- Student detail popup
- Performance report table
- PDF Export
- Responsive dashboard rendering

10.3.2 UI Test Cases Planned for Automation

No.	Automation Scenario
1	Verify dashboard loads successfully
2	Verify summary cards display correct metrics
3	Verify student search filters records correctly
4	Verify student detail popup displays complete evaluation
5	Verify PDF export is triggered successfully

10.4 AI Validation Automation

10.4.1 Automated AI Validations

- Verify score is within the valid range (0–100)
- Verify mandatory fields are populated
- Verify strengths and missing concepts are generated
- Verify recommendation is a single actionable sentence
- Verify next learning goal is present

10.4.2 AI Test Cases Planned for Automation

No.	Automation Scenario
1	Verify AI returns a valid JSON response with all mandatory fields.
2	Verify the generated score is within the valid range (0–100).
3	Verify strengths, missing concepts, and next learning goal are generated appropriately.
4	Verify the mentor recommendation is generated as a single actionable sentence.
5	Verify the AI response is successfully parsed into a valid StudentEvaluation object.

11. Defect Management

Priority	Severity	Description	Example
P1	Critical	Prevents core application functionality or blocks end-to-end evaluation.	Gemini API failure, application crash, dashboard not loading
P2	High	Major functionality affected, but application remains usable.	Incorrect student score, AI response parsing failure, PDF export failure
P3	Medium	Partial functionality affected with available workaround.	Incorrect analytics calculation, search filter not working, chart rendering issue
P4	Low	Cosmetic or usability issues with no impact on core functionality.	UI alignment, inconsistent labels, minor styling issues

12. Assumptions and Dependencies

12.1 Assumptions

ID	Assumption
A-01	The mentor summary accurately represents the topics covered during the classroom session.
A-02	The mentor summary is generated from a Tactiq.ai transcript with minimal manual modifications.
A-03	Student understanding is provided through exported WhatsApp conversations in the expected format.
A-04	Student names within the chat are sufficiently consistent to enable correct grouping of messages.
A-05	Google Gemini API is available throughout the evaluation process.
A-06	Gemini returns responses in the predefined JSON structure requested by the application.
A-07	Internet connectivity is available during AI evaluation.
A-08	The mentor reviews AI-generated insights before making academic decisions.

A-09	The application evaluates one classroom session at a time.
A-10	Input data is primarily English text.

12.2 Dependencies

ID	Dependency	Purpose
D-01	Google Gemini 2.5 Flash API	AI-powered student evaluation
D-02	Tactiq.ai	Generation of mentor reference transcript
D-03	WhatsApp Chat Export	Source of student discussions
D-04	Java 21	Backend application
D-05	Maven	Dependency and build management
D-06	React	Interactive mentor dashboard
D-07	Jackson	JSON serialization/deserialization
D-08	PDF Export Library	Report generation
D-09	Modern Web Browser	Dashboard access

13. Risks and Contingencies

13.1 Risks

Risk ID	Risk	Impact	Probability
R-01	AI service unavailable	High	Medium
R-02	AI response format changes	High	Medium
R-03	Poor quality mentor summary	High	Medium
R-04	Incomplete or inconsistent WhatsApp chat	Medium	Medium
R-05	Large classroom discussions increase evaluation time	Medium	Medium
R-06	Incorrect grouping of student messages due to inconsistent naming	Medium	Low
R-07	Network interruption during AI evaluation	High	Low
R-08	PDF export failure	Low	Low

13.2 Contingencies

Risk	Mitigation / Contingency
Gemini API unavailable	Display meaningful error message and allow the evaluation to be retried later.
Invalid AI response format	Validate the JSON response before parsing and terminate gracefully if invalid.
Poor mentor summary	Encourage review of the Tactiq.ai transcript before starting the evaluation.
Incomplete WhatsApp discussion	Evaluate only the available discussion and inform the mentor that results are based on the provided data.
Long AI response time	Display evaluation progress for each student to keep the user informed during processing.
Student naming inconsistencies	Encourage consistent naming conventions or verify grouped students before evaluation.
Internet connectivity issues	Retry the AI request or allow the evaluation to be restarted after connectivity is restored.

PDF generation failure	Preserve dashboard results and allow the user to retry PDF generation without repeating the evaluation.
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13.3 Limitations and Constraints

ID	Constraint
C-01	Supports text-based mentor summaries only.
C-02	Accepts pasted WhatsApp chats; direct WhatsApp integration is not available.
C-03	Evaluates one classroom session at a time.
C-04	AI evaluation depends on the quality of the mentor summary and student discussions.
C-05	Requires internet connectivity for Gemini API calls.
C-06	AI responses may vary slightly between executions due to the probabilistic nature of LLMs.

14. Test Deliverables

S.No.	Deliverable	Validation Focus
1	Input Data Validation	Mentor summary and WhatsApp chat processing
2	AI Evaluation Validation	Score generation, concept identification, recommendations
3	Analytics Validation	Class metrics, rankings, and performance calculations
4	Dashboard Validation	UI rendering, charts, search, and detailed student insights
5	Export Validation	JSON and PDF report generation
6	End-to-End System Validation	Complete workflow from input to final report

15. Execution Plan

Phase	Activity	ETA	Expected Outcome
Phase 1	Test Data Preparation	0.25 Day	Mentor summary and WhatsApp chat datasets prepared for testing
Phase 2	Functional Testing	2 Days	Core application features validated successfully
Phase 3	AI Accuracy Validation	2 Days	AI-generated evaluations verified for correctness and consistency
Phase 4	Negative & Edge Case Testing	1 Day	Application behaviour validated for invalid and boundary scenarios
Phase 5	End-to-End Testing	1 Day	Complete evaluation workflow verified from input to PDF export
Phase 6	Final Verification & Sign-off	0.5 Day	All critical test scenarios passed and application ready for submission

Total Estimated Testing Effort: 6.75 Days

16. Effort Estimation

Task Name	Ownership	Start Time	Finish Time	Status	Comments
Requirement & HLD Review	QAE-1	Day 1	Day 1	Completed	Reviewed business requirements and architecture to understand application flow
Test Planning	QAE-1	Day 1	Day 1	Completed	Prepared test strategy, scope, and testing approach
Test Data Preparation	QAE-1	Day 2	Day 2	Completed	Prepared mentor summary, WhatsApp chat datasets, and AI validation scenarios
Test Case Design	QAE-1	Day 2	Day 3	Completed	Created functional, AI, negative, edge, and integration test scenarios
Test Execution	QAE-1	Day 4	Day 5	Completed	Executed functional, AI accuracy, and end-to-end test cases
Defect Reporting & Tracking	QAE-1	Day 5	Day 6	Completed	Logged issues, tracked fixes, and verified resolutions
Regression Testing	QAE-1	Day 6	Day 7	Completed	Revalidated impacted modules after bug fixes
Test Report & Sign-off	QAE-1	Day 7	Day 7	Completed	Prepared Testing Evidence, execution summary, and QA sign-off

17. Open Questions